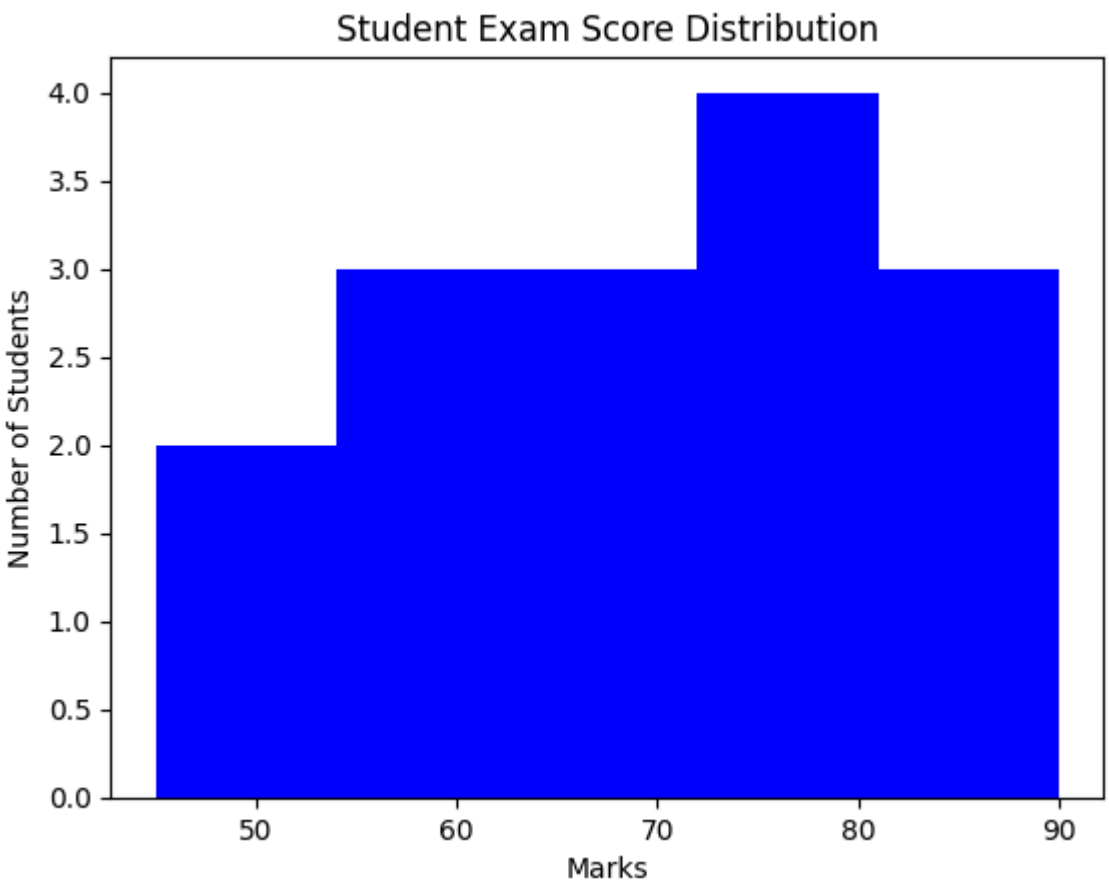
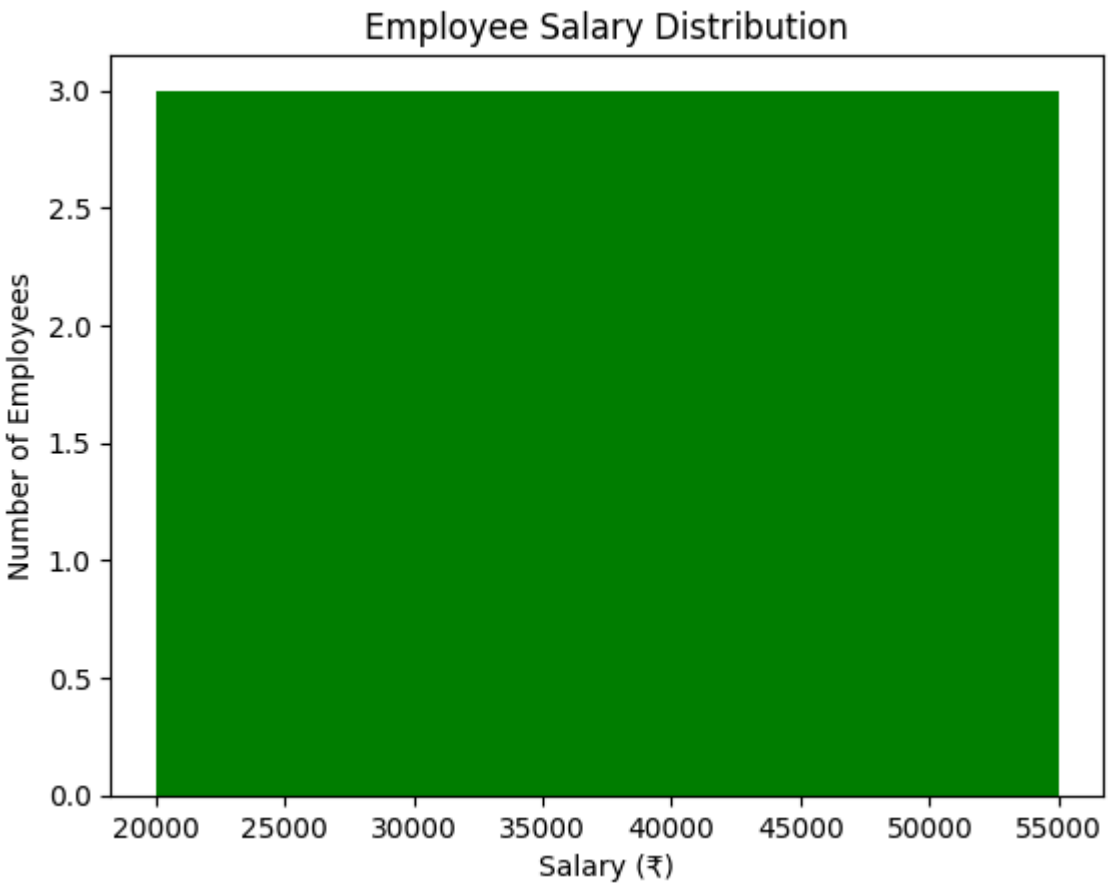
In [6]:
```
#  Program to create a histogram for Student Exam Scores

# Step 1: Import the matplotlib library and pyplot module for plotting graphs
import matplotlib.pyplot as plt

# Step 2: Create a list that stores marks obtained by students in an exam
# Unit of data = Marks out of 100
scores = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 83, 85, 90]

# Step 3: Create histogram
# bins=5 means the data will be divided into 5 groups
# color is used to give color to the bars of histogram
plt.hist(scores, bins=5, color="blue")

# Step 4: Add title to describe what the chart represents
plt.title("Student Exam Score Distribution")

# Step 5: Label for X-axis (represents marks)
plt.xlabel("Marks")

# Step 6: Label for Y-axis (represents number of students)
plt.ylabel("Number of Students")

# Step 7: Display the histogram
plt.show()
```



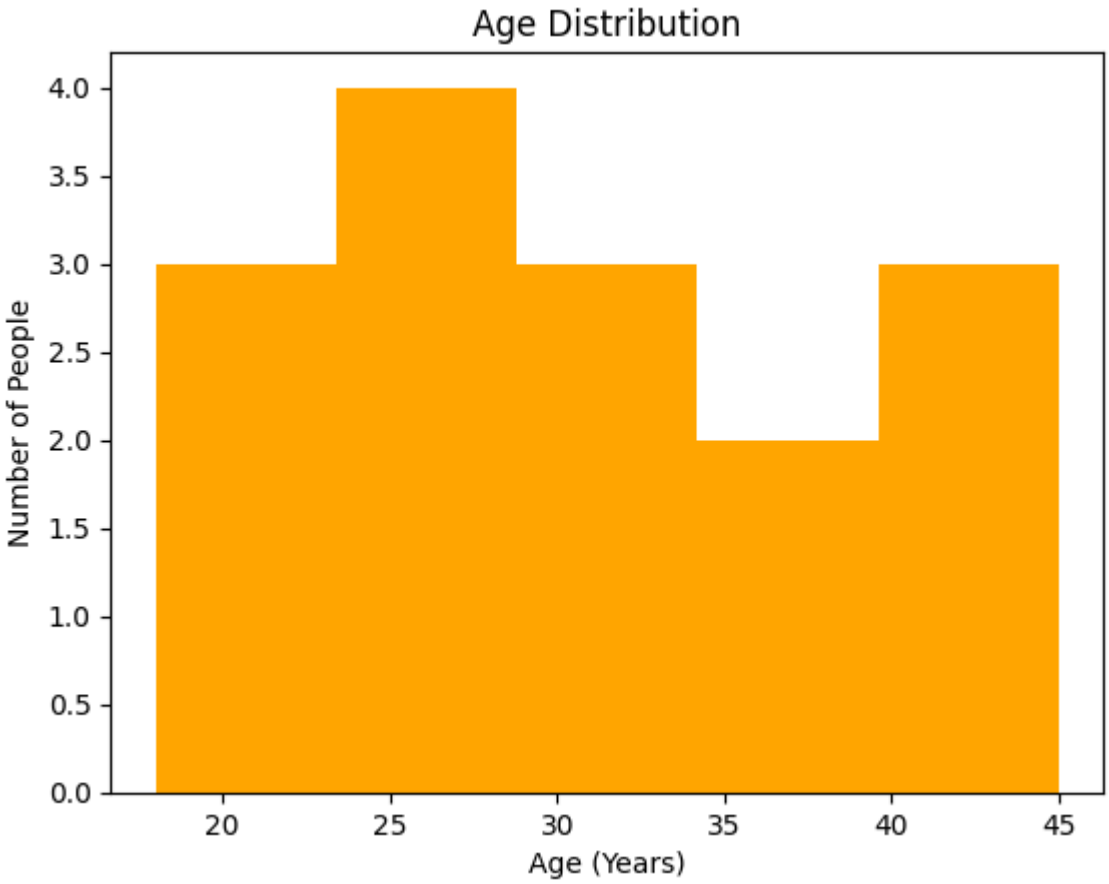
In [7]:
```
#  Program to create a histogram for Employee Salary Distribution

# Step 1: Create a list containing salary of employees
# Unit of data = Monthly Salary in Rupees
salary = [20000,22000,25000,27000,30000,32000,35000,37000,40000,42000,45000,4700
```

```
# Step 2: Create histogram to show distribution of salaries
# bins divide salaries into groups
plt.hist(salary, bins=5, color="green")

# Step 3: Add title for the graph
plt.title("Employee Salary Distribution")

# Step 4: Label X-axis to show salary values
plt.xlabel("Salary (₹)")

# Step 5: Label Y-axis to show number of employees
plt.ylabel("Number of Employees")

# Step 6: Display histogram
plt.show()
```

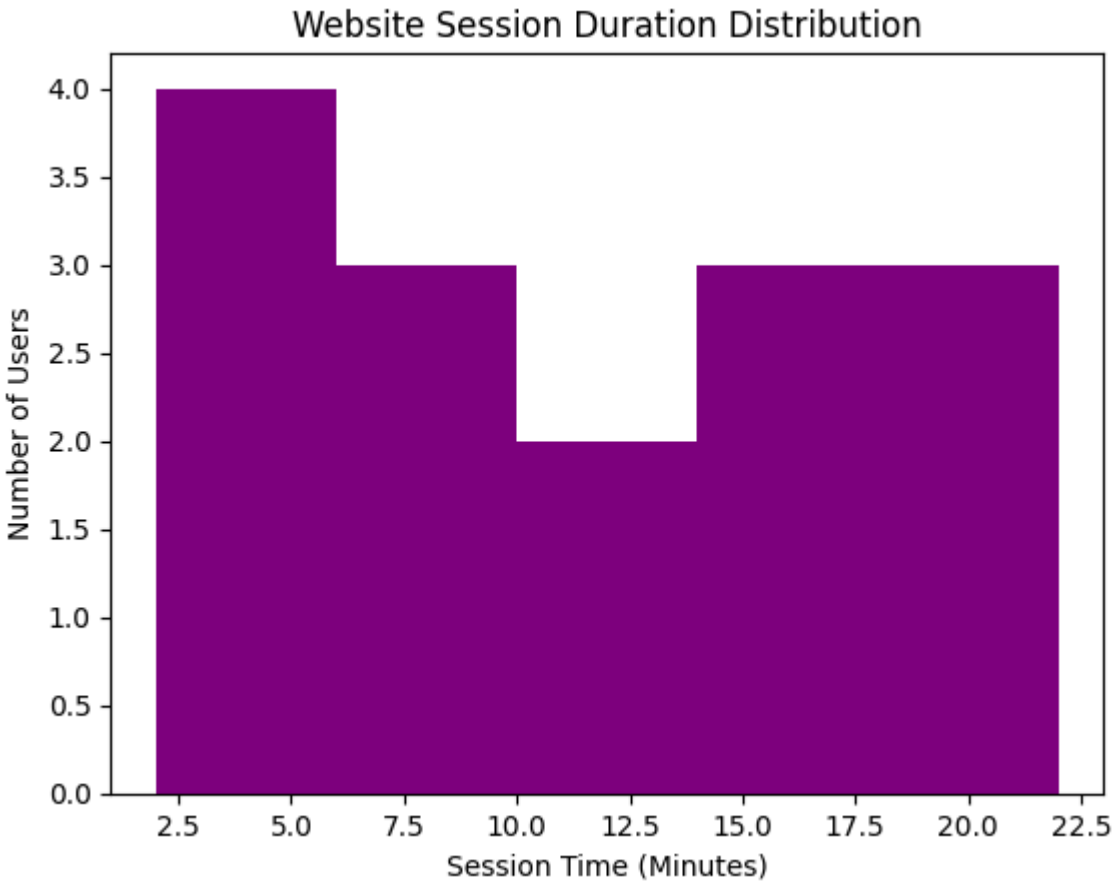


In [8]:
```
#  Program to create histogram for Age Distribution

# Step 1: List of ages of people
# Unit of data = Years
age = [18,20,22,24,25,27,28,30,32,34,35,37,40,42,45]

# Step 2: Create histogram
plt.hist(age, bins=5, color="orange")

# Step 3: Add title
plt.title("Age Distribution")

# Step 4: X-axis shows age
plt.xlabel("Age (Years)")
```

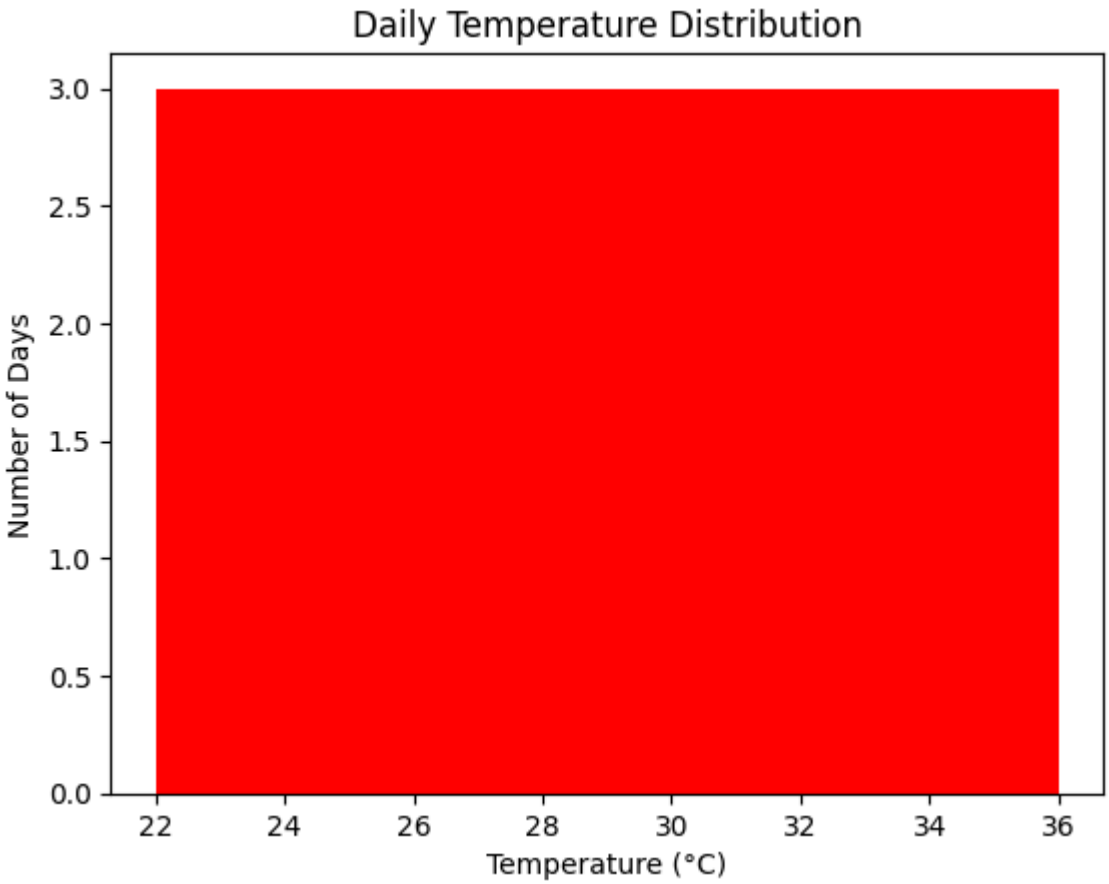
```
# Step 5: Y-axis shows number of people
plt.ylabel("Number of People")

# Step 6: Show histogram
plt.show()
```



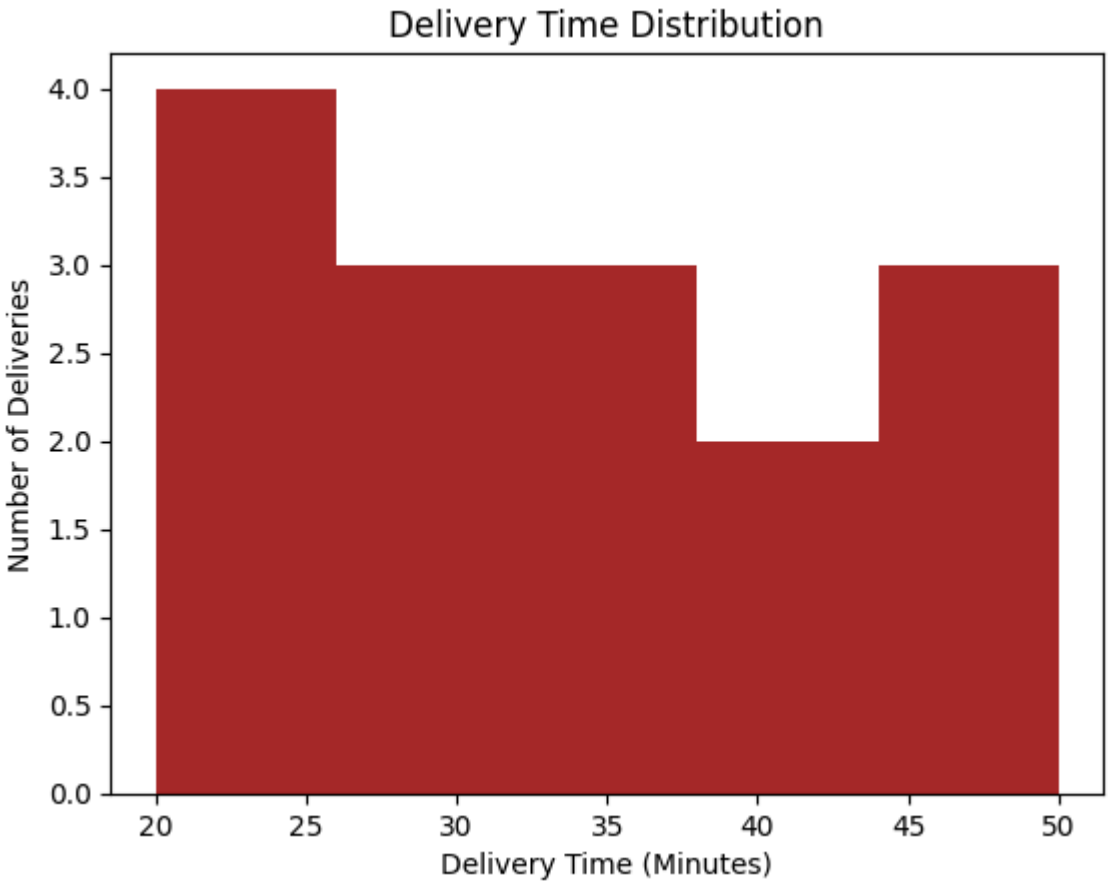
In [9]:
```
#  Program to create histogram for Website Session Duration

# Step 1: List containing duration of website sessions
# Unit of data = Minutes
session_time = [2,3,4,5,6,7,8,10,12,14,15,16,18,20,22]

# Step 2: Create histogram for session duration
plt.hist(session_time, bins=5, color="purple")

# Step 3: Title of the chart
plt.title("Website Session Duration Distribution")

# Step 4: Label X-axis
plt.xlabel("Session Time (Minutes)")

# Step 5: Label Y-axis
plt.ylabel("Number of Users")

# Step 6: Display histogram
plt.show()
```



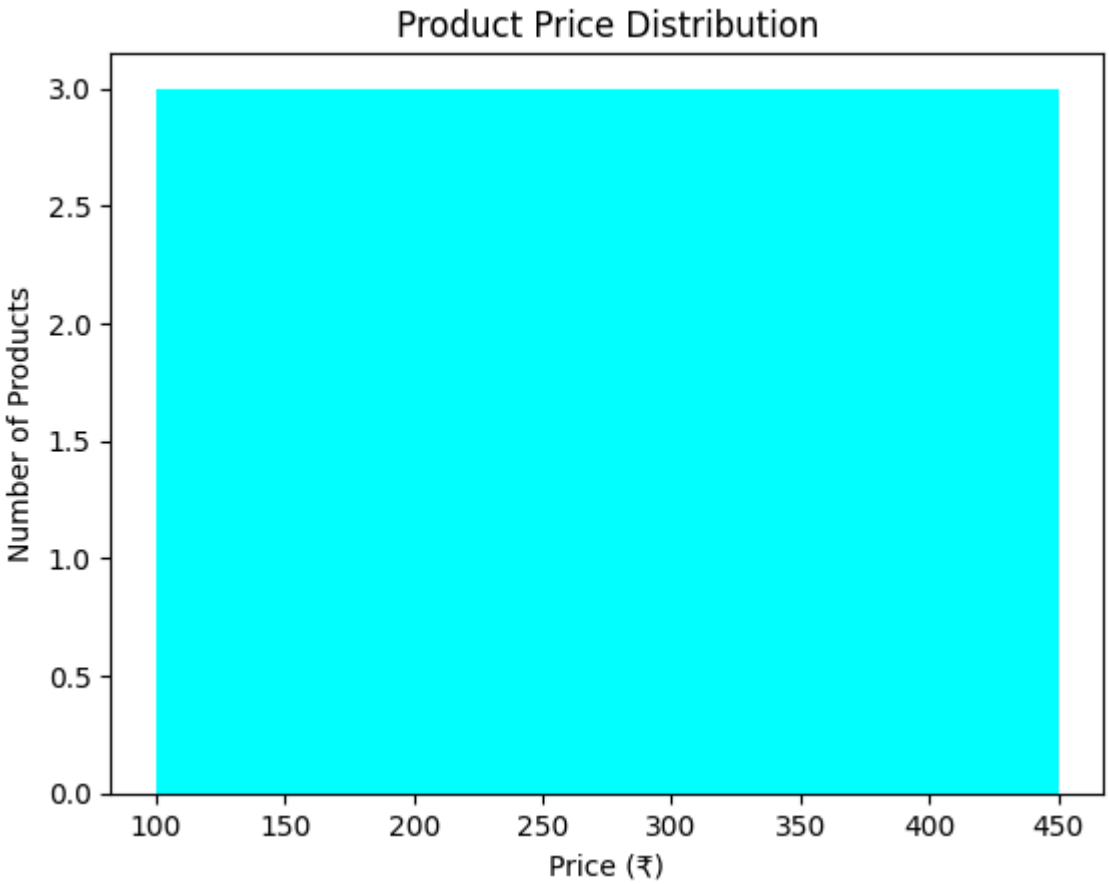
In [10]:
```
#  Program to create histogram for Daily Temperature

# Step 1: List containing temperature of different days
# Unit of data = Degree Celsius
temperature = [22,23,24,25,26,27,28,29,30,31,32,33,34,35,36]

# Step 2: Create histogram
plt.hist(temperature, bins=5, color="red")

# Step 3: Add title
plt.title("Daily Temperature Distribution")

# Step 4: Label X-axis
plt.xlabel("Temperature (°C)")

# Step 5: Label Y-axis
plt.ylabel("Number of Days")

# Step 6: Display the histogram
plt.show()
```



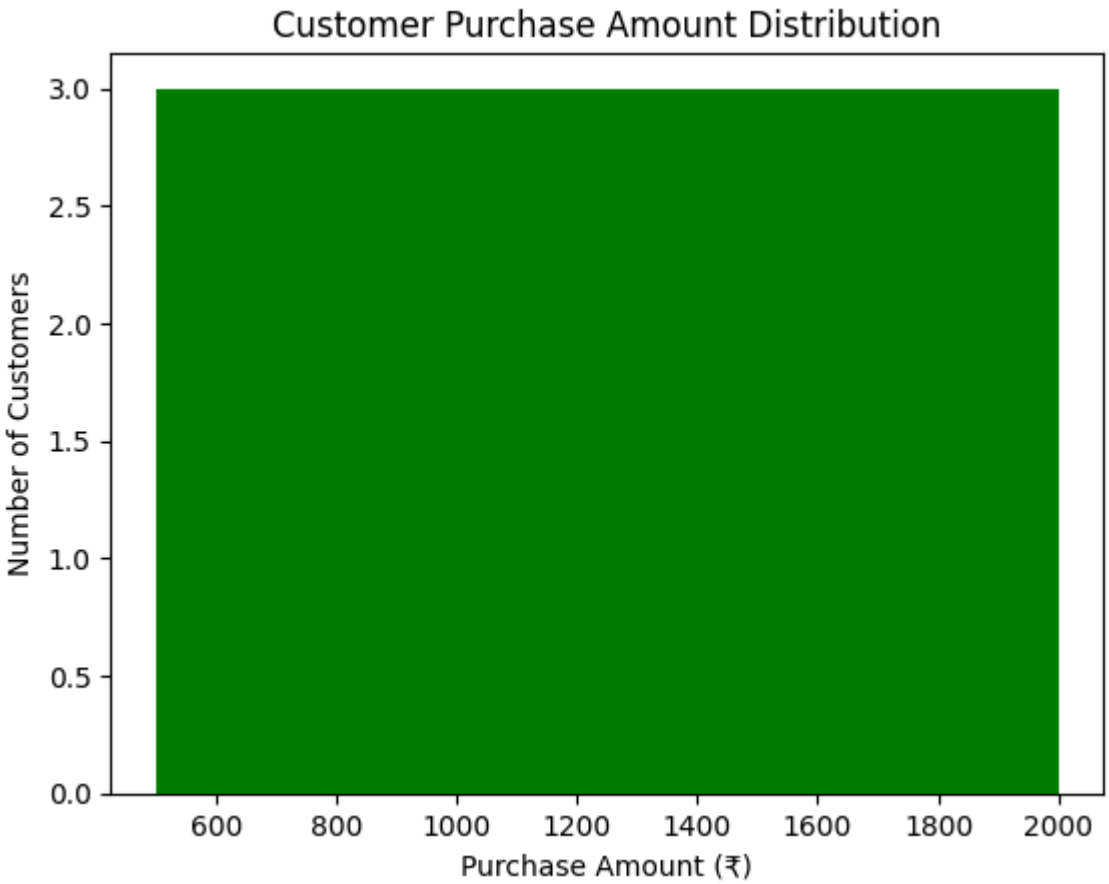
In [11]:
```
#  Program to create histogram for Delivery Time

# Step 1: List containing delivery times
# Unit of data = Minutes
delivery_time = [20,22,24,25,27,28,30,32,35,37,40,42,45,47,50]

# Step 2: Create histogram to show delivery time distribution
plt.hist(delivery_time, bins=5, color="brown")

# Step 3: Add title
plt.title("Delivery Time Distribution")

# Step 4: Label X-axis
plt.xlabel("Delivery Time (Minutes)")

# Step 5: Label Y-axis
plt.ylabel("Number of Deliveries")

# Step 6: Display histogram
plt.show()
```



In [12]:
```
# Program to create histogram for Product Prices

# Step 1: List of product prices
# Unit of data = Rupees
prices = [100,120,150,180,200,220,250,270,300,320,350,370,400,420,450]

# Step 2: Create histogram
plt.hist(prices, bins=5, color="cyan")

# Step 3: Add title
plt.title("Product Price Distribution")

# Step 4: Label X-axis
plt.xlabel("Price (₹)")

# Step 5: Label Y-axis
plt.ylabel("Number of Products")

# Step 6: Show histogram
plt.show()
```

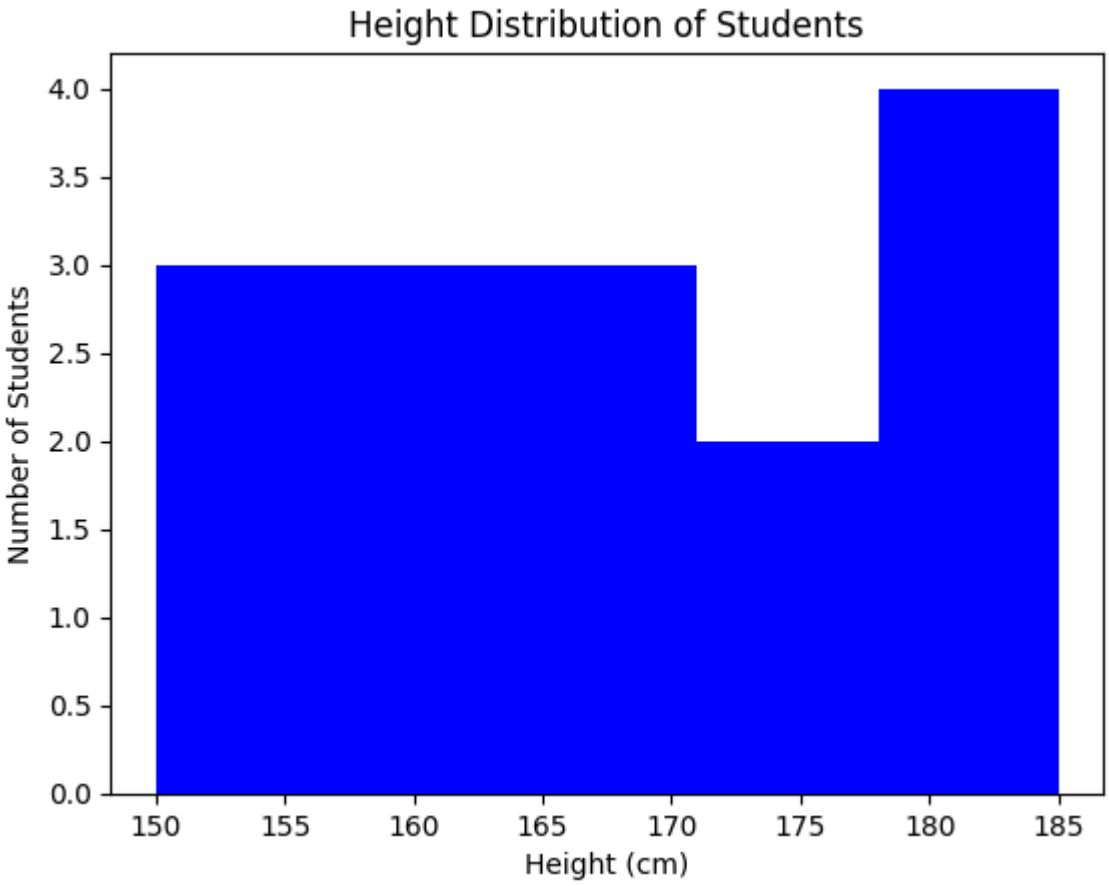


In [13]:
```
# Program to create histogram for Customer Purchase Amount

# Step 1: List containing purchase amount of customers
# Unit of data = Rupees
purchase_amount = [500,600,700,800,900,1000,1100,1200,1300,1400,1500,1600,1700,1

# Step 2: Create histogram
plt.hist(purchase_amount, bins=5, color="green")

# Step 3: Title of the chart
plt.title("Customer Purchase Amount Distribution")

# Step 4: Label X-axis
plt.xlabel("Purchase Amount (₹)")

# Step 5: Label Y-axis
plt.ylabel("Number of Customers")

# Step 6: Display histogram
plt.show()
```



In [14]:
```
#  Program to create histogram for House Rent Prices

# Step 1: List containing monthly house rent
# Unit of data = Rupees per month
rent = [5000,6000,7000,8000,9000,10000,11000,12000,13000,14000,15000,16000,17000

# Step 2: Create histogram
plt.hist(rent, bins=5, color="magenta")

# Step 3: Add title
plt.title("House Rent Price Distribution")

# Step 4: Label X-axis
plt.xlabel("Rent (₹ per month)")

# Step 5: Label Y-axis
plt.ylabel("Number of Houses")

# Step 6: Display histogram
plt.show()
```

In [15]:
```
# Program to create histogram for Heights of Students

# Step 1: List containing heights of students
# Unit of data = Centimeters
heights = [150,152,155,158,160,162,165,168,170,172,175,178,180,182,185]

# Step 2: Create histogram
plt.hist(heights, bins=5, color="blue")

# Step 3: Title of the chart
plt.title("Height Distribution of Students")

# Step 4: Label X-axis
plt.xlabel("Height (cm)")

# Step 5: Label Y-axis
plt.ylabel("Number of Students")

# Step 6: Display histogram
plt.show()
```


Height Distribution of Students
Number of Students
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
150
155
160
165
170
175
180
185
Height (cm)

In [ ]: